SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2024-2025

Program/s: B.Tech./MBA Tech/BTI

Year: III Semester: VI

Stream/s: All applicable streams

Subject: Elements of Project Management

No. of Pages 07

SYNOPTIC SOLUTION

Q1		Marks
a.	- Project management is define as the application of knowledge, skills, and tools necessary to achieve the project's requirements - Classification of project management in different environment 1. Construction Project Management 2. Manufacturing Project Management 3. IT Project Management 4. Biotechnology Project Management 5. Localization Project Management	2+2
b.	Parameters on which organization structure depends: 1. Rate of change of technology 2. Availability of resources 3. The product/service sold 4. Competition 5. Decision making requirements 6. Size of the Organization 7. Environment (legal, political, social and economic) 8. Culture (refers to the shared values, beliefs, customs, and practices that determine the behavior of members within a particular organization) [Any 4 to 5 parameters]	4
c.	List of Basic Processes of Project Planning: 1. Scope planning 2. Preparation of WBS- Work Breakdown Structures 3. Project Schedule Development & Staff planning 4. Resource planning 5. Budget planning 6. Procurement planning 7. Risk management 8. Quality planning 9. Communication planning	4
d.	- Risk management is an essential aspect of project management, and it plays a critical role in the success of any project - Risk management aims to identify, assess, and prioritize potential risks that could impact the project's objectives and then develop strategies to mitigate or avoid those risks -Project risk management help in responding to risk throughout the life of a project and in the best interests of meeting project objectives -Risk management is often overlooked in projects, but it can help improve project success by helping select good projects, determining project scope, and developing realistic estimates	4
e	Four ways of closing the project- 1. Closing by extinction 2. Closing by addition	2+2

	3. Closing by integration 4. Closing by starvation Brief explanation for any one	
Q2 a	-Systems approach is a way to visualize and analyze physical and conceptual systems, it is an approach for doing things—a framework for- ➢ Conceptualizing problems ➢ Solving problems ➢ Making decisions 1. Sub-System -A sub system is the operating level of a system. The parts that make up the structure of a system are called sub- systems -For example, the departments are a sub system of a company. A company in turn can be a sub system of a larger conglomerate 2. Environment -Each system is a part of environment, called "Supra System". -It includes all those factors external to a system that affect it and are generally uncontrollable -For example, the economic, legal, political and social factors that influence a business system are the environment of that business system 3. System Boundary -Each system has a boundary, i.e. outer limit of its operations and interaction. Open system: A system that dynamically interacts with its environment through a constant procurement of inputs and supply of outputs is an open system. Closed System: A closed system does not interact with its environment 4. Input-Output Conversion Process -A system involves input-output conversion process by importing the inputs from environment and transforming them through organizational and managerial processes adopted by sub systems, and exporting the outputs to the external environment 5. Weakness from Systems Approach -The change in one subsystem will affect other subsystems. But it does not explain the precise direction and magnitude of change - It does not offer a unified theory which can be applied to all types of organizations -It does not offer specific tools and techniques for the practicing executive	5X2 = 10Mark
Q2 b	Skills set / qualifications required for Project Manager: 1. Team Building 2. Leadership 3. Conflict resolution 4. Technical expertise 5. Planning 6. Organization 7. Entrepreneurship	5+5 = 10

	8. Administration 9. Management support 10. Resource allocation Explanation connecting these skills with success and failure of project	
Q3 a	A master plan is a long-term dynamic planning document that provides a conceptual framework to drive project development and growth OR Master Plan is a comprehensive document that outlines the key components, objectives, and strategies for successfully executing project (Definition: 2 Marks) Three Elements : 1. Scope, Charter or Statement of work(2Marks) 2. Management and Organization Section(2Marks) 3. Technical Section (4Marks) (List of three elements and explanation for each element)	10
Q3b	-Risk register is a document that contains the results of various risk management processes and that is often displayed in a table or spreadsheet format - A tool for documenting potential risk events and related information - *Contents of Risk register ▸ An identification number for each risk event ▸ A rank for each risk event ▸ The name of each risk event ▸ A description of each risk event ▸ The category under which each risk event falls ▸ The root cause of each risk ▪ Triggers for each risk; triggers are indicators or symptoms of actual risk events ▪ Potential responses to each risk ▪ The risk owner or person who will own or take responsibility for each risk ▪ The probability and impact of each risk occurring ▪ The status of each risk • The details of all qualitative and quantitative estimates made on risks, on states of the project's environment, or on project assumptions, complete with a brief description of the methods used to make such estimates • Minutes of all group meetings including all actions the group developed to deal with or mitigate each specific risk, including the decision to ignore a risk • The actual outcomes of identified risks and, if a risk came to occur, the results of actions taken to mitigate or transfer the risk or invoke the contingency plan Sample:	10

No.	Rank	Risk	Description	Category	Root Cause	Triggers	Potential Responses	Risk owner	Probability	Impacts	Status
R44	1										
R21	2										
R7	3										

Benefits of Risk register:

1. Identifies patterns from threats
2. Helps develop stronger risk mitigation strategies
3. Instills greater confidence in risk response
4. Contingency planning
5. Decision-making support
6. Documenting risks

(* At least 7 to 8 points in content list)

[Risk register 6 Marks + Benefits 4 Marks]

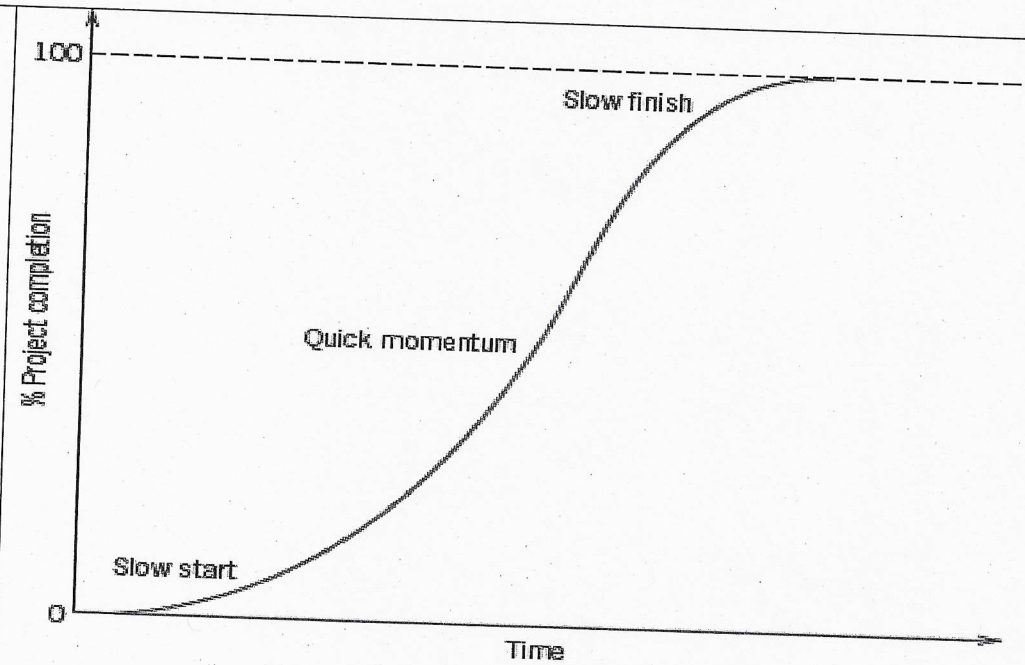
Q4 a	1. Sketch the network diagram – 03 Marks 2. critical path and total duration of the project -03 Marks 3. Determine Earliest start time, Earliest finish time, Latest start time and Latest finish time for each activity – 03 Marks 4. probability that the project duration will exceed 60 days? – 03 Marks	
Q4b	-The Delphi method was developed by Norman Dalkey and others at the RAND Corporation -This method is used to derive a consensus among a panel of experts who make predictions about future developments Generalized process of Delphi method- 1. Group of experts, make independent individual estimates 2. Estimates are used to compute group mean 3. Group mean is presented to the group 4. Group once again makes individual estimates 5. Mean is again found for individual estimates 6. If new mean is different from previous, then repeat step four 7. Otherwise Ê = Mean of final round of estimates -Provides independent and anonymous input regarding future events -Uses repeated rounds of questioning and written responses and avoids the biasing effects possible in oral methods, such as brainstorming -Application of Delphi method for risk identification (any example like in construction management, building big software) [Delphi Method: 5 Marks, Application:3Marks]	8
Q5 a	1. Project management (PM) allows us to accomplish more work in less time with fewer people 2. Profitability will increase 3. PM provides better control of scope changes 4. PM makes the organization more efficient and effective through better organizational behavior principles 5. PM allows to work more closely with customer	8

	6. PM provides a means for solving problems 7. PM enhances quality of project 8. PM helps to reduce power struggles 9. PM allows people to make good company decisions 10. PM delivers solutions 11. PM helps to increase business (At least 6 benefits with brief explanation: 8Marks)	
Q5b	-PMBOK defines Work Breakdown Structures as a deliverable –oriented grouping of activities that organizes and defines the total scope of project Factors need to be considered in Development of WBS: 1. Every activity in the WBS should produce a single tangible deliverable 2. Every activity at any level of the WBS is an aggregation of all its subordinate activities listed immediately below it 3. Each activity should be unique and distinct from other activities of the project 4. The activities should be decomposed logically from higher levels to lower levels 5. There should be some flexibility in the WBS development process, as the WBS might be updated when the project scope changes 6. The WBS must specify the important reporting points (e.g. review meetings, monthly reports, test report) 7. The activities should be compatible with organizational and accounting structures [Definition:02 Marks, Factors:06Marks]	8
Q5c	-A schedule is a timetable showing the forecast start and finish dates for activities or events within a Project, Programme or Portfolio -A project Schedule is at two levels - overall schedule and detailed schedule - Overall schedule comprises of major milestones and final date - Detailed schedule is the assignment of lowest level tasks to resources	4
Q6a	-A "qualitative risk" refers to an assessment of a potential risk based on subjective judgment and descriptive terms like "high," "medium," or "low" to evaluate the likelihood of the risk occurring and its potential impact, rather than using precise numerical data -Top Ten Risk Item Tracking is a qualitative risk analysis tool that helps to identify risks and maintain an awareness of risks throughout the life of a project -Establish a periodic review of the top ten project risk items -List the current ranking, previous ranking, number of times the risk appears on the list over a period of time, and a summary of progress made in resolving the risk item Example of Top Ten Risk Item Tracking:	8

Risk Event	Rank this month	Rank last month	Number of months in top ten	Risk resolution progress
Inadequate planning	1	2	4	Working on revising the entire project management plan
Poor definition	2	3	3	Holding meetings with project customer and sponsor to clarify scope
Absence of leadership	3	1	2	After previous project manager quit, assigned a new one to lead the project
Poor cost estimates	4	4	3	Revising cost estimates
Poor time estimates	5	5	3	Revising schedule estimates

[Definition:02 Marks, Top10 Risk items explanation:06 Marks]

Q. No.	Answer	Marks
Q6b	Process of project cost management- Step1: Resource Planning Step2: Cost Estimating Step3: Cost Budgeting Step4: Cost Control Brief explanation of any one step [List of steps: 2Marks, Brief explanation: 2Marks]	4
Q6c	1. Parametric Modeling [04Marks] 2. Computerized tools [04Marks]	8
Q7 a	- A unsuccessful project is a project when it fails to meet its established objectives on time, budget and performance List of key factors- 1. Probability of meeting technical objectives is very low 2. Inability of Research & Development to resolve the technical problems 3. Return on investment is not significant or very low 4. High cost involved in running it is an individual project 5. Market potential of the delivered output is very low 6. Constantly changing market needs 7. Requires a very long time to gain profits 8. Bears a negative impact on other projects 9. Problems created by intellectual property rights [Definition: 02Marks, Key factors (minimum six):06Marks]	8
Q7b	Factors that cause cost overrun- 1. Cost escalations 2. Time overruns 3. Scope changes 4. Estimation errors 5. Rectification and replacements 6. Unforeseen contingencies [Five Factors with brief description: 05 Marks, Sketch: 03 Marks]	8

Q7c	**Common stretched S-project life cycle** [Sketch:02Marks, Description:02Marks]	4